

Submit the following five exercises. You are expected to provide full solutions in a linear and coherent fashion. You are encouraged to work in groups, but your work must be your own and must be self-contained.

Exercise 1.

Find the domain of the function

$$f(x) = \frac{\sqrt{x-1}}{x^2 - x - 2}.$$

Exercise 2.

Find all real roots (zeros) of

$$g(x) = \begin{cases} x^2 + 1 & \text{if } x < 0, \\ \cos(x) & \text{if } 0 < x < 2, \\ -2x + 6 & \text{if } 2 < x. \end{cases}$$

Exercise 3.

For $0 \leq \theta < 2\pi$, find all real roots (zeros) of

$$h(x) = 6 \sin(3\theta) - 2\pi.$$

Exercise 4.

Find all real roots (zeros) of

$$i(x) = x^4 - 4x^2.$$

Exercise 5.

Find all real values of c such that

$$j(x) = \frac{x-1}{x^2 + 2cx + 4}$$

has domain all real numbers $\mathbb{R}$.